



Al-Balqa' Applied University / Faculty of Engineering Technology

General Chemistry / Mid-Term Exam

Time: 90 minutes

First Semester (2018/2019)

Date: 15/ 11/2018

Physics and Basic Sciences Dept.

Section: (1-1C)

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| Question No. | 1        | 2        | 3        | 4        | 5        | 6        | 7        | 8         | 9        | 10       |
|--------------|----------|----------|----------|----------|----------|----------|----------|-----------|----------|----------|
| Answer       | <u>a</u> | <u>d</u> | <u>b</u> | <u>c</u> | <u>b</u> | <u>a</u> | <u>c</u> | <u>CA</u> | <u>e</u> | <u>d</u> |
| Question No. | 11       | 12       | 13       | 14       | 15       | 16       | 17       | 18        | 19       | 20       |
| Answer       | <u>b</u> | <u>e</u> | <u>a</u> | <u>e</u> | <u>e</u> | <u>b</u> | <u>a</u> | <u>Bd</u> | <u>a</u> | <u>e</u> |

Useful Data: Avogadro's No. =  $6.022 \times 10^{23}$

Q1: Circle the most correct answer in the following questions:

1. The set of quantum numbers,  $n = 4$ ,  $l = 3$ ,  $m_l = 2$  is describing an electron in a...

- (a) 4f orbital      b. 3p orbital      c. it is not allowed      d. 4d orbital      e. 4s orbital

2. Which of the following sets of quantum numbers is not allowed?

- a.  $n = 3$ ,  $l = 1$ ,  $m_l = -1$ ,  $m_s = +\frac{1}{2}$       b.  $n = 2$ ,  $l = 1$ ,  $m_l = 0$ ,  $m_s = +\frac{1}{2}$   
 c.  $n = 3$ ,  $l = 0$ ,  $m_l = 0$ ,  $m_s = +\frac{1}{2}$       d.  $n = 2$ ,  $l = 2$ ,  $m_l = -1$ ,  $m_s = -\frac{1}{2}$       e. all are allowed

3. How many moles of hydrogen atoms exists in 0.01 mole of  $(\text{NH}_4)_2\text{HPO}_4$ ?

- a. 0.01      b. 0.09      c. 0.5      d. 0.08      e. 0.02

4. How many grams of  $\text{N}_2\text{O}$  are formed, if 0.46 mole of  $\text{NH}_4\text{NO}_3$  is used in the reaction? Given the unbalanced chemical equation.  $\text{NH}_4\text{NO}_3 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$

- a. 40 g      b. 80 g      c. 20 g      d. 10 g      e. 16g

5. Consider the reaction:  $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$

If 0.86 mole of  $\text{MnO}_2$  and 48.2 g of HCl react, how many grams of  $\text{Cl}_2$  will be produced?

- a. 46.8      b. 11.7      c. 64.2      d. 18.5      e. 23.4

6. Lithium reacts with nitrogen when heated to form lithium nitride according to the following balanced equation:  $6\text{Li}(s) + \text{N}_2(g) \rightarrow 2\text{Li}_3\text{N}(s)$

If 5.89 g of  $\text{Li}_3\text{N}$  is formed when 12.3 g of Li are heated with 33.6 g of  $\text{N}_2$ , what is the percent yield of the reaction?      a. 14.3 %      b. 48.2%      c. 34.2%      d. 28.6%      e. 57.2%

7. What is the percent water in hydrated calcium chloride  $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ ?

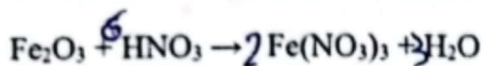
- a. 66.67%      b. 32.47%      c. 24.51%      d. 12.26%      e. 16.23%

28  
40

المجموع = 28

14  
2  
28

8. Calculate the minimum sum of all coefficients in the **balanced** chemical reaction:

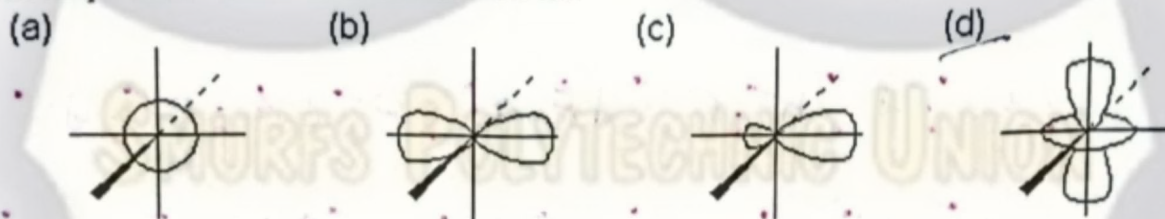


- a. 12      b. 8      c. 6      d. 15      e. 18

9. Which of the following ions has the smallest diameter?

- a.  $\text{O}^{2-}$     b.  $\text{Na}^+$     c.  $\text{F}^-$     d.  $\text{Ne}$     e.  $\text{Al}^{3+}$

10. The shape of one of the atomic 3d orbitals is...



11. Choose the electron configuration of the zinc ( $_{30}\text{Zn}$ ) atom (in its normal state):

- a.  $[\text{Ar}]4s^2 4p^6$     b.  $[\text{Ar}]4s^2 3d^{10}$     c.  $[\text{Ar}]4s^1 4d^{10}$     d.  $[\text{Kr}]5s^2 5d^{10}$     e.  $[\text{Kr}]5s^1 5d^{10}$

12. How many orbitals are there on the 3<sup>rd</sup> energetic level?

- a. 16    b. 2    c. 3    d. 4    e. 9

13. Atomic radii of chemical elements in a **group** **increase** from:

- a. left to right    b. right to left    c. top to bottom    d. bottom to top.    e. stays constant

14. Which of the following electron configurations represents the electron configuration for a magnesium cation  $_{12}\text{Mg}^{2+}$ ?

- a.  $1s^2 2s^2 2p^6 3s^2$     b.  $1s^2 2s^2 2p^6 3s^2 3p^2$     c.  $1s^2 2s^2 2p^4$     d.  $1s^2 2s^2 2p^5$     e.  $1s^2 2s^2 2p^6$

15. Which of the following species has the **most** unpaired electrons  $\text{S}^+$ ,  $\text{S}^{2+}$ ,  $\text{S}$ , or  $\text{S}^-$ ?

- a.  $\text{S}$     b.  $\text{S}^-$     c.  $\text{S}^+$     d.  $\text{S}^{2+}$     e. all have the same number of unpaired electrons.

16. A substance (X) containing C, H, N, and O. Given the following percent composition by mass: 19.8% C, 2.5% H, 11.6% N. What is the molecular formula given that the mass of 0.01 mole of X is weighing 1.21 g?

- a.  $\text{C}_4\text{H}_6\text{N}_2\text{O}_{10}$     b.  $\text{C}_2\text{H}_3\text{NO}_5$     c.  $\text{C}_3\text{H}_3\text{NO}_4$     d.  $\text{C}_4\text{H}_3\text{N}_2\text{O}_5$     e.  $\text{C}_5\text{H}_5\text{NO}_3$

17. Calculate the number of N, in  $1.68 \times 10^4$  g of urea  $[(\text{NH}_2)_2\text{CO}]$ .

- a.  $3.37 \times 10^{26}$     b.  $3.37 \times 10^{23}$     c.  $6.74 \times 10^{26}$     d.  $6.7 \times 10^{25}$     e.  $1.19 \times 10^{26}$

18. Which element of the following has the highest electron affinity?

- a.  $_{20}\text{Ca}$     b.  $_{35}\text{Br}$     c.  $_{33}\text{As}$     d.  $_{34}\text{Se}$     e.  $_{36}\text{Kr}$

19. Which property is identical for elements from the same period?

- a. atomic radius    b. ionization energy    c. the number of valence electron  
d. chemical properties    e. None is correct

20. Which element would have the **greatest difference** between the first ionization energy ( $I_1$ ) and the second ionization energy ( $I_2$ )? a.  $_{19}\text{K}$     b.  $_{20}\text{Ca}$     c.  $_{13}\text{Al}$     d.  $_{14}\text{Si}$     e.  $_{3}\text{Li}$